LOCAL MULTIBUBBLE AND CASCADE EXCLUSION FOR TYPE II NAVIER–STOKES BLOWUP

Abstract. We prove the local multibubble and cascade reduction used in the Type II Navier–Stokes exclusion argument. The setting is a positive local Caffarelli–Kohn–Nirenberg concentration sequence at a possible singular point. All arguments are local in parabolic cylinders. Compact-cylinder failure, splitting of active CKN mass, strict same-point scale cascades, separated active physical points, and localized gauge degeneracy are reduced by explicit active-frame decoupling. Once exterior profiles and remainders are shown to be perturbative in every selected active frame, the remaining branch is either single-core, scale-rigid, or incompatible with the local Type II condition. Thus no local multibubble, cascade, or gauge-degenerate Type II configuration remains. This paper uses the classical theory of suitable weak solutions of the Navier–Stokes equations, including Leray [Ler34], Caffarelli–Kohn–Nirenberg [CKN82], Scheffer [Sch76], and Lin [Lin98].

1. Local setting and active frames

Let (u_n, p_n) be parabolic blowups around a possible singular point with positive local Caffarelli–Kohn–Nirenberg concentration. On

$$Q_R = B_R \times (-R^2, 0),$$

write

$$A(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |u_n(y,s)|^2 \, dy,$$

$$E(R) = R^{-1} \int_{Q_R} |\nabla u_n|^2 \, dy \, ds,$$

$$C(R) = R^{-2} \int_{Q_R} |u_n|^3 \, dy \, ds, \qquad D(R) = R^{-2} \int_{-R^2}^{0} \int_{B_R} |p_n - (p_n)_{B_R}(s)|^{3/2} \, dy \, ds.$$

Pressure is always normalized modulo functions of time.

A local active frame is a sequence of centers and scales (x_n, λ_n), $\lambda_n > 0$, with associated critical rescaling

$$\mathcal{T}_{x_n,\lambda_n} f(y) = \lambda_n f(x_n + \lambda_n y).$$

If a profile $\phi \in L^3_\sigma(\mathbb{R}^3)$ is observed in the frame (x_n, λ_n), its contribution has the form

$$\phi_n(y) = \lambda_n \phi(x_n + \lambda_n y).$$

The transformation is an isometry on $L^3(\mathbb{R}^3)$:

$$\|\mathcal{T}_{x_n,\lambda_n} f\|_{L^3(\mathbb{R}^3)} = \|f\|_{L^3(\mathbb{R}^3)}.$$

Definition 1.1 (Bounded selected-window limit)**.** A selected-window subsequence has a bounded selected-window limit on a compact cylinder Q_R if the corresponding velocities converge strongly in $L^3(Q_R)$ to a limit in $L^\infty(Q_R)$. The limit is nonzero if it is not identically zero on Q_R.

Definition 1.2 (Local Type II sequence)**.** A positive local concentration sequence is called a local Type II sequence if no selected-window rescaling has a nonzero bounded selected-window limit on any compact cylinder.

Definition 1.3 (Local multibubble/cascade branch)**.** A local multibubble/cascade branch is a positive-concentration local Type II branch for which at least one of the following structural alternatives occurs:

(i) the compact-cylinder suitable package $A+E+C+D$ is unbounded on some fixed rescaled cylinder;
(ii) positive CKN mass splits into two or more comparable active cores;
(iii) a strict same-point scale cascade appears inside bounded rescaled cylinders;
(iv) separated physical points remain active in different local rescaled frames;
(v) the localized scale or centering selection degenerates because no unique core is selected.

Definition 1.4 (Active profile package)**.** An active profile package is a finite or countable family

$$\{(\phi^j, x_n^j, \lambda_n^j)\}_{j\in J}, \qquad \phi^j \in L^3_\sigma(\mathbb{R}^3),$$

obtained along a positive local concentration sequence, with pairwise orthogonal critical parameters after comparable parameters at the same physical point have been grouped. A profile is active if its critical mass is bounded from below by a fixed number $\eta_* > 0$:

$$\|\phi^j\|_{L^3(\mathbb{R}^3)} \ge \eta_*.$$

The package has finite critical mass if

$$\sum_{j\in J} \|\phi^j\|^3_{L^3(\mathbb{R}^3)} \le M_* < \infty.$$

Lemma 1.5 (Finite active profile count)**.** *Every active profile package with finite critical mass contains only finitely many active profiles. More precisely,*

$$\#J \le M_* \eta_*^{-3}.$$

Proof. For every active profile,

$$\|\phi^j\|^3_{L^3} \ge \eta_*^3.$$

If $J_0 \subset J$ is finite, then

$$\#J_0\, \eta_*^3 \le \sum_{j\in J_0} \|\phi^j\|^3_{L^3} \le M_*.$$

Taking the supremum over finite subsets J_0 gives

$$\#J \le M_* \eta_*^{-3}.$$

Thus J is finite. □

2. Elementary local decoupling estimates

Lemma 2.1 (Critical rescalings that escape are locally invisible)**.** *Let $\phi \in L^3(\mathbb{R}^3)$, let $R < \infty$, and define*

$$W_n(y) = \rho_n \phi(z_n + \rho_n y),$$

where $\rho_n > 0$. Assume that the sets

$$z_n + \rho_n B_R$$

either have measure tending to zero, or escape to spatial infinity in the sense that for every compact set $K \subset \mathbb{R}^3$,

$$(z_n + \rho_n B_R) \cap K = \varnothing$$

for all sufficiently large n. Then

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. By the critical change of variables,

$$\|W_n\|_{L^3(B_R)}^3 = \int_{B_R} \rho_n^3 |\phi(z_n + \rho_n y)|^3 \, dy = \int_{z_n + \rho_n B_R} |\phi(x)|^3 \, dx.$$

Fix $\varepsilon > 0$. If $|z_n + \rho_n B_R| \to 0$, then absolute continuity of the integral gives

$$\int_{z_n + \rho_n B_R} |\phi|^3 < \varepsilon$$

for all sufficiently large n. If the sets escape to infinity, choose a compact set $K \subset \mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

For all sufficiently large n, the escaping assumption gives $(z_n + \rho_n B_R) \cap K = \varnothing$, and therefore

$$\|W_n\|_{L^3(B_R)}^3 \le \int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon$$

for all sufficiently large n. Since ε is arbitrary, the local L^3 norm tends to zero. □

Lemma 2.2 (Separated active profiles vanish in the selected frame)**.** *Let* (x_n^p, λ_n^p) *and* (x_n^q, λ_n^q) *be two active frames with separated physical centers in the sense that*

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

If $\phi^q \in L^3(\mathbb{R}^3)$*, then the* q*-profile observed in the* p*-frame converges to zero locally in* L^3*: for every* $R < \infty$*,*

$$\left\| \frac{\lambda_n^p}{\lambda_n^q} \phi^q \left(\frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} y \right) \right\|_{L^3(B_R)} \to 0.$$

Proof. The displayed function has the form in Theorem 2.1 with

$$z_n = \frac{x_n^p - x_n^q}{\lambda_n^q}, \qquad \rho_n = \frac{\lambda_n^p}{\lambda_n^q}.$$

The corresponding physical set in the q-profile variables is

$$z_n + \rho_n B_R = \frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} B_R.$$

The separation condition says that, in the p-frame, the q-center leaves every bounded set. Indeed, the distance from the origin to the set

$$z_n + \rho_n B_R$$

is bounded below by

$$\frac{|x_n^q - x_n^p|}{\lambda_n^q} - R\frac{\lambda_n^p}{\lambda_n^q} = \frac{\lambda_n^p}{\lambda_n^q} \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right).$$

For fixed R, the parenthesis tends to $+\infty$. Thus either the sets escape every compact subset of the q-profile variables, or their measure tends to zero along a further subsequence. In both cases Theorem 2.1 gives the claim. □

Lemma 2.3 (Strict outer scales vanish in the innermost frame)**.** *Let several active profiles have the same physical center and scales* $\lambda_n^1, \dots, \lambda_n^J$*. Suppose* λ_n^1 *is an innermost scale and*

$$\frac{\lambda_n^1}{\lambda_n^j} \to 0 \qquad (j \neq 1).$$

Then every outer profile $\phi^j \in L^3(\mathbb{R}^3)$, viewed in the λ_n^1-frame, converges to zero in $L^3(B_R)$ for each fixed $R < \infty$. Any constant ambient velocity arising from the regular part of the outer flow is absorbed into the translation modulation.

Proof. In the innermost variables, the j-th outer contribution has the form

$$W_n^j(y) = \frac{\lambda_n^1}{\lambda_n^j}\phi^j\left(\frac{\lambda_n^1}{\lambda_n^j}y + o(1)\right).$$

Let $\rho_n = \lambda_n^1/\lambda_n^j$. By the critical change of variables,

$$\|W_n^j\|_{L^3(B_R)}^3 = \int_{\rho_n B_R + o(1)} |\phi^j(x)|^3 \, dx.$$

Since $|\rho_n B_R| \to 0$, absolute continuity of the integral gives

$$\|W_n^j\|_{L^3(B_R)} \to 0$$

for every $\phi^j \in L^3$. Equivalently, for a general $\phi^j \in L^3$, choose a smooth compactly supported approximation ϕ_ε^j in L^3. The critical scaling is an L^3-isometry, so the error produced by $\phi^j - \phi_\varepsilon^j$ is bounded by

$$\|\phi^j - \phi_\varepsilon^j\|_{L^3},$$

uniformly in n, and the compactly supported approximation has vanishing local norm by the preceding shrinking-set argument. Letting first $n \to \infty$ and then $\varepsilon \downarrow 0$ gives the same conclusion. A constant vector field from the regular part of an outer flow contributes only a Galilean drift in the selected frame; it is therefore incorporated into the translation modulation and does not remain as an exterior perturbation. □

Lemma 2.4 (Compactness failure produces another core)**.** *If the compact-cylinder suitable package is unbounded on a fixed rescaled cylinder, then after passing to a subsequence the local $C + D$ quantity is unbounded on a comparable cylinder. Consequently the branch contains an additional local CKN concentration core.*

Proof. Fix radii $0 < r < R < \infty$. The local Caccioppoli inequality for suitable weak solutions gives a constant $C_{r,R}$ such that

$$A(r) + E(r) \le C_{r,R}\big(1 + C(R)^{2/3} + C(R) + D(R)^{2/3}\big)$$

for the rescaled suitable pair, after the pressure is normalized by subtracting spatial means. Therefore, if $C(R) + D(R)$ were bounded on every comparable larger cylinder, then $A(r) + E(r)$ would also be bounded on the smaller cylinder. Adding the assumed boundedness of $C(r) + D(r)$ would give a bounded compact-cylinder suitable package on Q_r. Taking the contrapositive, unboundedness of $A + E + C + D$ on a fixed cylinder forces unboundedness of $C + D$ on a comparable cylinder.

Choose radii and a subsequence so that

$$C(R) + D(R) \to \infty.$$

By absolute continuity in space-time and the positivity of the CKN density, one can select a smaller parabolic subcylinder where a fixed positive amount of $C + D$ is present. Rescaling around that subcylinder gives another local CKN concentration core. This is the asserted additional core. □

3. Terminal active-frame decoupling

Definition 3.1 (Admissible terminal active package)**.** Fix a compact cylinder $B_{2R} \times I$ and a selected active frame. A terminal active package is admissible on this cylinder if the following hold.

(i) The active profiles satisfy the finite-mass and active-mass conditions of Theorem 1.4.

(ii) Every nonselected active profile is either separated from the selected physical point in the sense of Theorem 2.2, or has the same physical point and a strictly outer scale in the sense of Theorem 2.3, after comparable same-point scales have been grouped.

(iii) The rescaled remainder r_n satisfies

$$\|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0,$$

and its linear profile-evolution residual tends to zero in

$$L^1_I H^{-1}(B_R).$$

(iv) The selected velocity satisfies

$$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

(v) The selected modulation coefficients satisfy

$$\sup_n \big(\|a_n\|_{L^\infty(I)} + \|b_n\|_{L^\infty(I)}\big) < \infty.$$

(vi) Local pressure decomposition is stable under the splitting: if tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$, then the corresponding pressure gradients tend to zero in $L^1_I H^{-1}(B_R)$, after subtracting spatial means.

(vii) Cutoff commutators generated by localizing to B_R are controlled by the $L^\infty_I L^3$, $L^1_I L^{3/2}$, and pressure-gradient norms appearing above.

Theorem 3.2 (Terminal profile-evolution decoupling)**.** *Consider an admissible terminal active package on $B_{2R} \times I$. Let U_n be the selected-frame velocity and let S_n be the sum, in that frame, of all nonselected active profiles together with the rescaled remainder. Then on $B_R \times I$:*

(i) $\nabla \cdot S_n = 0$ *distributionally;*

(ii) $S_n \to 0$ *in* $L^\infty_I L^3(B_{2R})$*;*

(iii) $U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0$ *in* $L^1_I L^{3/2}(B_R)$*;*

(iv)
$$\partial_\tau S_n - \nu \Delta S_n + a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n \to 0 \quad \textit{in } L^1_I H^{-1}(B_R);$$

(v) *every pressure source containing at least one factor of S_n has vanishing pressure gradient in $L^1_I H^{-1}(B_R)$;*

(vi) *localization errors introduced by compact cutoffs vanish in $L^1_I H^{-1}(B_R)$;*

(vii) *the effective modulation parameters of the selected frame remain bounded;*

(viii) *after discarding S_n, the selected frame remains a positive finite-mass local Type II branch with pressure reconstruction and the compact-cylinder bounds required by the single-core analysis.*

Proof. Each profile in the package is divergence-free, and the rescaled remainder is divergence-free. Since divergence commutes with the Navier–Stokes critical rescalings, finite sums remain divergence-free in distributions. This proves (i).

We prove (ii). By admissibility, every nonselected active profile is either separated from the selected physical point or is a same-point strictly outer scale. The separated profiles vanish locally by Theorem 2.2, and the same-point outer profiles vanish locally by Theorem 2.3. The remainder tends to zero in $L^\infty_I L^3(B_{2R})$ by Theorem 3.1. The number of active profiles is finite by Theorem 1.5. Since every summand in S_n tends to zero in $L^\infty_I L^3(B_{2R})$, the triangle inequality gives

$$\|S_n\|_{L^\infty_I L^3(B_{2R})} \le \sum_{j\ne j_*} \|S^j_n\|_{L^\infty_I L^3(B_{2R})} + \|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0.$$

This proves (ii).

For (iii), admissibility gives

$$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

Hence, by Hölder's inequality,

$$\|U_n \otimes S_n\|_{L^1_I L^{3/2}(B_R)} \le |I| \|U_n\|_{L^\infty_I L^3(B_R)} \|S_n\|_{L^\infty_I L^3(B_R)} \to 0,$$

and the same estimate applies to $S_n \otimes U_n$. Finally,

$$\|S_n \otimes S_n\|_{L^1_I L^{3/2}(B_R)} \le |I| \|S_n\|^2_{L^\infty_I L^3(B_R)} \to 0.$$

For (iv), write the represented Navier–Stokes equation for the full profile sum $U_n + S_n$ and subtract the represented equation for the selected component U_n. The remaining equation for S_n has the displayed linear left-hand side and forcing terms consisting of nonlinear stresses from (iii), profile-evolution residuals of the remainder, and cutoff errors. The nonlinear stresses tend to zero in $L^1_I L^{3/2}(B_R)$, hence in $L^1_I H^{-1}(B_R)$, because $L^{3/2}(B_R) \hookrightarrow H^{-1}(B_R)$ on a bounded three-dimensional ball. The residual term vanishes by Theorem 3.1. The cutoff terms vanish by the localization part of admissibility, recorded explicitly in (vi). This proves the displayed convergence in $L^1_I H^{-1}(B_R)$.

For (v), decompose the pressure source of $U_n + S_n$ into the pure selected source, the pure exterior source, and the two mixed sources. The mixed sources are

$$\partial_i\partial_j(U_{n,i}S_{n,j} + S_{n,i}U_{n,j}) \quad \text{and} \quad \partial_i\partial_j(S_{n,i}S_{n,j}).$$

By (iii), their tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$. The pressure-stability part of admissibility, equivalently local Calderon–Zygmund estimates plus harmonic pressure control modulo spatial means, gives convergence of the corresponding pressure gradients to zero in $L^1_I H^{-1}(B_R)$. Pure exterior sources are generated by profiles that are locally invisible in the selected frame; applying the same pressure stability after Theorems 2.2 and 2.3 gives their vanishing pressure gradients.

For (vi), let χ_R be a cutoff equal to one on B_R and supported in B_{2R}. The commutators are finite sums of terms of the following types:

$$(\nabla\chi_R)\cdot\nabla S_n, \qquad (\Delta\chi_R)S_n, \qquad (\nabla\chi_R)\cdot(U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n),$$

plus analogous pressure cutoff terms. Their supports lie in the fixed annulus $B_{2R}\setminus B_R$. By the localization part of admissibility, these commutators are controlled by the norms already shown to vanish in (ii), (iii), and (v). Hence all localization errors vanish in $L^1_I H^{-1}(B_R)$.

For (vii), the only exterior contribution that can remain nonzero at first order in a compact selected frame is a constant ambient velocity. This has already been absorbed into the translation modulation. The remaining exterior profiles vanish locally, so their contribution to the modulation equations is small in the same local L^3 topology used above. The selected modulation coefficients are bounded by admissibility; hence the effective parameters remain bounded.

For $(viii)$, the positive CKN mass of the selected profile is unchanged by subtracting a term that tends to zero in local L^3 and whose mixed pressure sources vanish by (v). The pressure reconstruction is stable under the same decomposition by the pressure-stability condition in Theorem 3.1. The compact-cylinder bounds pass to the selected component because the full package is bounded and the discarded terms vanish on compact cylinders. Thus the selected frame remains a positive finite-mass local Type II branch with the analytic inputs needed for the single-core analysis. □

4. Same-point and separated-point reductions

Theorem 4.1 (Same-point cascade reduction)**.** *Every same-point multibubble or strict same-point cascade reduces, in the selected innermost or compound frame, to one of the following: the single-core criterion, scale-rigidity, or a perturbative terminal branch covered by Theorem 3.2, provided the terminal package generated in the selected innermost frame is admissible in the sense of Theorem 3.1.*

Proof. Partition the active same-point profiles into scale-comparability classes. If two scales satisfy

$$0 < c \leq \frac{\lambda_n^i}{\lambda_n^j} \leq C < \infty,$$

then, after passing to a subsequence, they are represented in one common active frame. Their sum is again divergence-free. If the compound profile has critical norm below the small-data threshold, it is perturbative and is removed from the active list. If it remains active, then for the purpose of local selection it is one compound core rather than several independent bubbles.

If the resulting compound core is unique and the localized scale-centering Jacobian is nondegenerate, the branch is a single selected core. The local single-core criterion applies to this branch. If the derivative is degenerate, then the local scale or center is not uniquely determined by the selected mass; this is precisely the gauge-degenerate alternative in Theorem 1.3.

It remains to consider strict same-point scale separation. Choose an innermost active scale. Every outer profile is locally invisible in that innermost frame by Theorem 2.3, except for constant drifts, which are absorbed into the translation modulation. Therefore the sum of all outer profiles and the remainder is the perturbative term S_n of Theorem 3.2. The selected innermost branch is then governed, up to a vanishing L^1H^{-1} error, by the same represented equation as a single active profile. If it has the vanishing-dissipation compact structure, the single-core criterion excludes it. If the limiting scale equation produces a nonzero bounded selected-window limit, the local Type II condition excludes it. Otherwise the branch is the terminal perturbative case covered by Theorem 3.2. These alternatives exhaust same-point multibubbles and strict same-point cascades. □

Theorem 4.2 (Separated-point decoupling)**.** *Separated active physical points are locally invisible in each other's active rescaled frames. Consequently every separated-point multibubble reduces in each selected active frame to a single-core, scale-rigid, or perturbative terminal branch, provided the terminal package in that selected active frame is admissible in the sense of Theorem 3.1.*

Proof. Fix an active point p with frame (x_n^p, λ_n^p). Let $q \neq p$ be another active point. By separation,

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

Thus Theorem 2.2 gives local L^3 convergence to zero of the q-profile in the p-frame on every compact ball. The same conclusion holds for each profile centered at any physical point different from p. Since only finitely many active profiles remain after the critical mass threshold is imposed, the sum of all profiles outside the p-frame tends to zero locally in L^3.

The pure pressure generated by a separated profile is locally invisible by the same change of variables whenever its profile pressure belongs to $L^{3/2}_{\mathrm{loc}}$ with the standard Calderon–Zygmund control. Mixed pressure terms contain at least one exterior velocity factor, hence vanish in

$L^1 H^{-1}_{\mathrm{loc}}$ by the pressure part of Theorem 3.2. Localization errors and modulation effects are the remaining perturbative errors handled in Theorem 3.2.

Therefore, in the selected p-frame, all other physical points contribute only a perturbative forcing tending to zero in $L^1 H^{-1}$ on compact cylinders. The selected frame remains a positive finite-mass branch with the same compact-cylinder and pressure reconstruction properties. Hence the branch reduces to the single-core case, the scale-rigid case, or the perturbative terminal branch. Since p was arbitrary, the same reduction holds in every active frame. □

5. Multibubble and cascade exclusion

Assumption 5.1 (Local single-core exclusion)**.** Every compact single-core branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable package, bounded modulation, and vanishing localized dissipation is excluded from the local Type II regime.

Assumption 5.2 (Local scale-rigidity exclusion)**.** Every scale-rigid local branch produced by the same-point or separated-point reductions either has a nonzero bounded selected-window limit or reduces to the single-core branch in Theorem 5.1. Consequently no such branch is compatible with the local Type II condition.

Assumption 5.3 (Terminal admissibility in active frames)**.** Every terminal package produced by the same-point and separated-point reductions is admissible on each compact selected cylinder in the sense of Theorem 3.1.

Theorem 5.4 (Local multibubble and cascade exclusion)**.** *Assume the local single-core and scale-rigidity exclusions Theorems 5.1 and 5.2, and assume terminal admissibility in active frames, Theorem 5.3. Then no local multibubble, cascade, or gauge-degenerate Type II case can occur in the class described above.*

Proof. Let a local Type II branch be in the multibubble/cascade class of Theorem 1.3. If the compact-cylinder suitable package fails, Theorem 2.4 produces an additional local concentration core, so the branch is reduced to active mass splitting, same-point cascading, or separated active points.

After passing to a subsequence, every pair of active cores has one of three relations: comparable scale at the same physical point, strictly separated scale at the same physical point, or separated physical center. Comparable same-point cores are grouped into compound profiles. A perturbative compound profile is removed by small-data stability; an active compound profile is a single selected core if its scale-centering selection is nondegenerate. If the selection is degenerate, the branch is gauge-degenerate, and the same grouping shows that the failure is non-uniqueness of the selected active core rather than a new independent local dynamics.

Strict same-point scale cascades are handled by Theorem 4.1. In the innermost frame, all outer scales are locally invisible after constant drifts are absorbed. The branch therefore reduces to the single-core criterion, the scale-rigidity criterion, or terminal perturbative decoupling. Separated active physical points are handled by Theorem 4.2; in each active frame, all other physical points vanish locally and only perturbative pressure and localization errors remain.

The single-core alternative is excluded by Theorem 5.1. The scale-rigid alternative is excluded by Theorem 5.2. The terminal perturbative alternative cannot sustain an independent multibubble obstruction because Theorem 3.2 removes all exterior profiles and remainders from the selected active equation in $L^1 H^{-1}_{\mathrm{loc}}$, preserves the positive finite-mass selected branch, and leaves only the single-core or scale-rigid possibilities already excluded. Consequently every alternative in Theorem 1.3 is impossible in the local Type II regime. □

Corollary 5.5 (Removal of the multibubble alternative)**.** *In the local Type II analysis, every failure of the single nondegenerate compact core alternative is excluded by Theorem 5.4 under the hypotheses stated there.*

Proof. By Theorem 1.3, failure of the single nondegenerate compact core alternative means that at least one of the following occurs: the compact-cylinder suitable package is unbounded; positive local CKN mass splits between active cores; a strict same-point scale cascade occurs; separated physical points remain active in distinct local frames; or the localized scale or center selection is degenerate. The compactness-failure case produces an additional core by Theorem 2.4. Same-point and separated cases are reduced by Theorems 4.1 and 4.2. The final exclusion is Theorem 5.4. Hence no failure of the single nondegenerate compact core alternative remains. □

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